

# Smart Blood Donor Network System

## Problem

Finding blood donors during emergencies is difficult.

## Solution

Connect hospitals, blood banks, and voluntary donors.

## Features

- Donor registration
- Blood group search
- Emergency alerts
- Hospital requests
- Donation history
- Eligibility reminders

## Additional Features:

### AI-Based Donor Availability Prediction

Using historical data like:

- Previous donations
- Frequency of accepting requests
- Response times
- Availability records

We can predict the **"likelihood of the donor responding"**

Output:

- High probability (90%)
- Medium probability (60%)
- Low probability (20%)

Simple models can be used like:

- Logistic Regression

- Random Forest
- XGBoost

### Intelligent Donor Recommendation Engine

Instead of showing all matching donors, we calculate a **Donor Priority Score**.

Factors:

- Blood group match
- Distance from hospital/patient
- Last donation date
- Donor availability status
- Number of previous successful donations
- Response rate to previous requests

Example:

| Donor | Distance | Last Donation | Score |
|-------|----------|---------------|-------|
| A     | 3 km     | 150 days ago  | 92    |
| B     | 1 km     | 95 days ago   | 85    |
| C     | 8 km     | 200 days ago  | 80    |

This ranks donors automatically.

## **Analytics Dashboard**

Admin dashboard showing:

- Most requested blood groups
- Monthly donation trends
- Response rates
- Hospital-wise requests
- Donor retention rate
- Demand vs Supply graphs